

2024 Higher Chemistry Paper 2 - Q3

Section: Nature's Chemistry

Topic: Alcohols, Aldehydes and Ketones

Question summary (Q3):

Describe the experimental procedures and expected results to distinguish between alcohols, aldehydes and ketones.

Worked Solution:

- Distinguishing alcohols:
 - Use acidified potassium dichromate solution (orange → green if oxidation occurs).
 - Primary alcohols oxidise to aldehydes → colour change.
 - Secondary alcohols oxidise to ketones → colour change.
 - Tertiary alcohols are not easily oxidised → no colour change.
- Distinguishing aldehydes vs ketones:
 - Use Tollens' reagent (ammoniacal silver nitrate).
 - Aldehydes reduce Tollens' → silver mirror forms.
 - Ketones: no reaction.
- Alternative: Fehling's solution (blue copper(II) complex).
- Aldehydes reduce → red precipitate of Cu_2O .
- Ketones: no reaction.

Final Answer:

- Alcohols tested with acidified dichromate: primary & secondary oxidise (orange → green), tertiary do not.
- Aldehydes give positive Tollens' (silver mirror) or Fehling's (red precipitate).
- Ketones show no reaction with these reagents.

Revision Tips:

- Primary alcohol → aldehyde (then further to carboxylic acid).
- Secondary alcohol → ketone.

- Tertiary alcohol → resistant to oxidation.
- Aldehydes are easily oxidised; ketones are not.
- Remember: Tollens' = silver mirror; Fehling's = red precipitate.